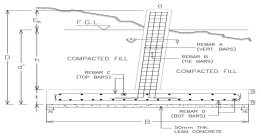
FOUNDATION SCHEDULE

	STRUCTURE NO.	TOWER TYPE	SOIL TYPE	STRUCTURE DIMENSION						REINFORCEMENT STEEL BARS										BILL OF QUANTITIES PER LEG				
				PAD			CHIMNEY			REBAR "A"		REBAR "B"		REBAR "C"		REBAR "D"				EXCAVATION	LEAN CONCRETE	REBAR	CONCRETE	BACKFILL
				B	t	d'	h	Eh	a	BAR DIA.	NO. OF PCS.	BAR DIA.	NO. OF PCS.	BAR DIA.	NO. OF PCS. EACH WAY	BAR DIA.	NO. OF PCS. EACH WAY			CU.M.	CU.M.	KGS	CU.M.	CU.M.
1	LA-03	DU2	III	6300	700	4000	3300	1000	1000	20	36	12	23	16	38	16	52			160.74	1.98	2,597.82	32.08	127.68
2	LA-04	DR2St	II	4100	550	4000	3450	1000	1000	20	32	12	23	16	18	16	25			68.08	0.84	1,128.56	13.70	54.54
3	LA-05	DR2St	III	4400	550	4000	3450	1000	1000	20	32	12	23	16	19	16	28			78.41	0.97	1,233.02	15.10	63.34
4	LA-06	DS2	III	4500	600	4000	3400	1000	1000	20	32	12	23	16	21	16	28			82.01	1.01	1,299.06	16.55	65.45
5	LA-07	DU2	III	6300	700	4000	3300	1000	1000	20	36	12	23	16	38	16	52			160.74	1.98	2,597.82	32.08	127.68
6	LA-08	DU2	IV	6500	700	4000	3300	1000	1000	20	36	12	23	16	46	16	54			171.11	2.11	2,889.75	33.88	136.13
7	LA-09	DU2 Leg A & B	III	6300	700	4000	3300	1000	1000	20	36	12	23	16	38	16	52			160.74	1.98	2,597.82	32.08	127.68
		DU2 Leg C & D	III	6000	900	4200	3300	4000	1000	32	52	12	38	16	44	16	55			153.00	1.80	5,212.15	39.70	115.50
8	LA-10	DU2 Leg A, C & D	II	5900	700	4000	3300	1000	1000	20	36	12	23	16	33	16	48			140.98	1.74	2,289.63	28.67	111.57
		DU2 LEG B	II	5600	900	4200	3300	4000	1000	32	52	12	38	16	41	16	51			133.28	1.57	4,933.64	35.52	100.19
9	LA-11	DU2 Leg A & B	II	5900	700	4000	3300	1000	1000	20	36	12	23	16	33	16	48			140.98	1.74	2,289.63	28.67	111.57
		DU2 Leg C & D	II	5600	900	4200	3300	4000	1000	32	52	12	38	16	41	16	51			133.28	1.57	4,933.64	35.52	100.19
10	LA-12	DU2	II	5900	800	4100	3300	2500	1000	20	60	12	31	16	38	16	49			144.46	1.74	3,052.15	33.65	111.57
11	LA-13	DS2 Leg A, C & D	III	4500	600	4000	3400	1000	1000	20	32	12	23	16	21	16	28			82.01	1.01	1,299.06	16.55	65.45
		DS2 Leg B	III	4300	750	4150	3400	4000	1000	32	28	12	38	16	15	16	33			77.66	0.92	2,428.93	21.27	59.47
12	LA-14	DR2St	III	4400	550	4000	3450	1000	1000	20	32	12	23	16	19	16	28			78.41	0.97	1,233.02	15.10	63.34
13	LA-18	DU2	II	5900	700	4000	3300	1000	1000	20	36	12	23	16	33	16	48			140.98	1.74	2,289.63	28.67	111.57
14	LA-17	DR2St	III	4400	550	4000	3450	1000	1000	20	32	12	23	16	19	16	28			78.41	0.97	1,233.02	15.10	63.34
15	LA-21	DT4/DU4	III	6500	850	4150	3300	1000	1100	20	48	12	23	16	38	16	60			177.45	2.11	3,127.23	41.12	135.43
16	LB-02	DU2	III	6300	700	4000	3300	1000	1000	20	36	12	23	16	38	16	52			160.74	1.98	2,597.82	32.08	127.68
17	LB-03	DR2st	II	4100	550	4000	3450	1000	1000	20	32	12	23	16	18	16	25			68.08	0.84	1,128.56	13.70	54.54
18	LB-07	DS2	IVA	6200	600	4000	3400	1000	1000	20	32	12	23	16	39	16	41			155.68	1.92	2,241.92	27.46	127.30
19	LB-08	DS2	II	4200	600	4000	3400	1000	1000	20	32	12	23	16	20	16	26			71.44	0.88	1,205.80	14.98	56.58
20	LB-09	DU2	II	5900	700	4000	3300	1000	1000	20	36	12	23	16	33	16	48			140.98	1.74	2,289.63	28.67	111.57
21	LB-10	DS2	II	4200	600	4000	3400	1000	1000	20	32	12	23	16	20	16	26			71.44	0.88	1,205.80	14.98	56.58
22	LB-11	DS2	II	4200	600	4000	3400	1000	1000	20	32	12	23	16	20	16	26			71.44	0.88	1,205.80	14.98	56.58
23	LB-12	DR2st	II	4100	550	4000	3450	1000	1000	20	32	12	23	16	18	16	25			68.08	0.84	1,128.56	13.70	54.54
24	LB-13	DS2	III	4500	600	4000	3400	1000	1000	20	32	12	23	16	21	16	28			82.01	1.01	1,299.06	16.55	65.45
26	LB-15	DR2st	III	4400	550	4000	3450	1000	1000	20	32	12	23	16	19	16	28			78.41	0.97	1,233.02	15.10	63.34
26	LB-16	DR2st	III	4400	550	4000	3450	1000	1000	20	32	12	23	16	19	16	28			78.41	0.97	1,233.02	15.10	63.34
27	LB-17	DR2st	IV	4500	550	4000	3450	1000	1000	20	32	12	23	16	19	16	29			82.01	1.01	1,264.11	15.59	66.41
28	LB-19	DR2st	IV	4500	550	4000	3450	1000	1000	20	32	12	23	16	19	16	29			82.01	1.01	1,264.11	15.59	66.41
29	LB-22	DS2	III	4500	600	4000	3400	1000	1000	20	32	12	23	16	21	16	28			82.01	1.01	1,299.06	16.55	65.45
30	LB-23	DT4/DU4	III	6500	850	4150	3300	1000	1100	20	48	12	23	16	38	16	60			177.45	2.11	3,127.23	41.12	135.43
																		Subtotal		3,832.96	46.84	72,389.11	811.14	3,022.90
																		Total		13,300.01 final	162.96 final	233,813.55 final	2,757.01 final	10,525.73 final
																		Contract volume		15,260.00	189.00	219,943.00	2,999.00	12,182.00
																		Difference		1,959.99	26.04	- 13,870.55	241.99	1,656.28
																				1.15	1.16	0.94	1.09	1.16
																		Unit Cost		1,200.00	11,000.00	115.00	14,200.00	500.00
																		Total Cost		18,312,000.00	2,079,000.00	25,293,445.00	42,585,800.00	6,091,000.00
																								94,361,245.00
																		Total Cost		15,960,010.82	1,792,510.50	26,888,558.68	39,149,520.70	5,292,882.50
																								89,053,463.18
																				2,351,989.20	286,489.50	- 1,595,113.68	3,436,279.30	828,137.50
																								- 5,307,781.82



REBAR CALCULATION

	Stub Setting			REINFORCEMENT STEEL BARS (CUTTING LIST)																						
				REBAR "A"					REBAR "B"				REBAR "C"						REBAR "D"							
				BAR	BAR	L_A	NO. OF	WEIGHT	BAR	NO. OF	L_B	WEIGHT	BAR	L_C	c	NO. OF PCS.	WEIGHT		BAR	L_C	c	NO. OF PCS.	WEIGHT			
	Rad.	dec. deg.	deg	L sloping	L vertical	MM	PCS.	KG	L	PCS.	MM	KG	L	MM	MM	TOTAL	KG		L	MM	MM	TOTAL	KG			
1	1.35	77.40	77°24'16"	5.229	5.110	4929	36	464.18	3.640	23	850	74.34	7.250	6150	550	76	869.48		7.250	6150	550	104	1,189.81			
2	1.39	79.58	79°34'57"	5.191	5.110	4891	32	409.60	3.640	23	850	74.34	4.750	3950	400	36	269.84		4.750	3950	400	50	374.78			
3	1.39	79.58	79°34'57"	5.191	5.110	4891	32	409.60	3.640	23	850	74.34	5.050	4250	400	38	302.82		5.050	4250	400	56	446.26			
4	1.35	77.25	77°15'1"	5.232	5.110	4932	32	412.84	3.640	23	850	74.34	5.250	4350	450	42	347.95		5.250	4350	450	56	463.93			
5	1.35	77.40	77°24'16"	5.229	5.110	4929	36	464.18	3.640	23	850	74.34	7.250	6150	550	76	869.48		7.250	6150	550	104	1,189.81			
6	1.35	77.40	77°24'16"	5.229	5.110	4929	36	464.18	3.640	23	850	74.34	7.450	6350	550	92	1,081.56		7.450	6350	550	108	1,269.66			
7	1.35	77.40	77°24'16"	5.229	5.110	4929	36	464.18	3.640	23	850	74.34	7.250	6150	550	76	869.48		7.250	6150	550	104	1,189.81			
	1.35	77.40	77°24'16"	8.508	8.310	8208	52	2,792.86	3.640	38	850	122.83	7.350	5850	750	88	1,020.65		7.350	5850	750	110	1,275.81			
8	1.35	77.40	77°24'16"	5.229	5.110	4929	36	464.18	3.640	23	850	74.34	6.850	5750	550	66	713.41		6.850	5750	550	96	1,037.69			
	1.35	77.40	77°24'16"	8.508	8.310	8208	52	2,792.86	3.640	38	850	122.83	6.950	5450	750	82	899.30		6.950	5450	750	102	1,118.64			
9	1.35	77.40	77°24'16"	5.229	5.110	4929	36	464.18	3.640	23	850	74.34	6.850	5750	550	66	713.41		6.850	5750	550	96	1,037.69			
	1.35	77.40	77°24'16"	8.508	8.310	8208	52	2,792.86	3.640	38	850	122.83	6.950	5450	750	82	899.30		6.950	5450	750	102	1,118.64			
10	1.35	77.40	77°24'16"	6.868	6.710	6568	60	1,016.22	3.640	31	850	100.20	7.050	5750	650	76	845.49		7.050	5750	650	98	1,090.24			
11	1.35	77.25	77°15'1"	5.232	5.110	4932	32	412.84	3.640	23	850	74.34	5.250	4350	450	42	347.95		5.250	4350	450	56	463.93			
	1.35	77.25	77°15'1"	8.461	8.260	8161	28	1,495.64	3.640	38	850	122.83	5.350	4150	600	30	253.27		5.350	4150	600	66	557.19			
12	1.39	79.58	79°34'57"	5.191	5.110	4891	32	409.60	3.640	23	850	74.34	5.050	4250	400	38	302.82		5.050	4250	400	56	446.26			
13	1.35	77.40	77°24'16"	5.229	5.110	4929	36	464.18	3.640	23	850	74.34	6.850	5750	550	66	713.41		6.850	5750	550	96	1,037.69			
14	1.39	79.58	79°34'57"	5.191	5.110	4891	32	409.60	3.640	23	850	74.34	5.050	4250	400	38	302.82		5.050	4250	400	56	446.26			
15	1.28	73.53	73°32'2"	5.472	5.260	5172	48	647.74	4.040	23	950	82.51	7.750	6350	700	76	929.44		7.750	6350	700	120	1,467.54			
16	1.35	77.40	77°24'16"	5.229	5.110	4929	36	464.18	3.640	23	850	74.34	7.250	6150	550	76	869.48		7.250	6150	550	104	1,189.81			
17	1.39	79.58	79°34'57"	5.191	5.110	4891	32	409.60	3.640	23	850	74.34	4.750	3950	400	36	269.84		4.750	3950	400	50	374.78			
18	1.35	77.25	77°15'1"	5.232	5.110	4932	32	412.84	3.640	23	850	74.34	6.950	6050	450	78	855.43		6.950	6050	450	82	899.30			
19	1.35	77.25	77°15'1"	5.232	5.110	4932	32	412.84	3.640	23	850	74.34	4.950	4050	450	40	312.44		4.950	4050	450	52	406.18			
20	1.35	77.40	77°24'16"	5.229	5.110	4929	36	464.18	3.640	23	850	74.34	6.850	5750	550	66	713.41		6.850	5750	550	96	1,037.69			
21	1.35	77.25	77°15'1"	5.232	5.110	4932	32	412.84	3.640	23	850	74.34	4.950	4050	450	40	312.44		4.950	4050	450	52	406.18			
22	1.35	77.25	77°15'1"	5.232	5.110	4932	32	412.84	3.640	23	850	74.34	4.950	4050	450	40	312.44		4.950	4050	450	52	406.18			
23	1.39	79.58	79°34'57"	5.191	5.110	4891	32	409.60	3.640	23	850	74.34	4.750	3950	400	36	269.84		4.750	3950	400	50	374.78			
24	1.35	77.25	77°15'1"	5.232	5.110	4932	32	412.84	3.640	23	850	74.34	5.250	4350	450	42	347.95		5.250	4350	450	56	463.93			
26	1.39	79.58	79°34'57"	5.191	5.110	4891	32	409.60	3.640	23	850	74.34	5.050	4250	400	38	302.82		5.050	4250	400	56	446.26			
26	1.39	79.58	79°34'57"	5.191	5.110	4891	32	409.60	3.640	23	850	74.34	5.050	4250	400	38	302.82		5.050	4250	400	56	446.26			
27	1.39	79.58	79°34'57"	5.191	5.110	4891	32	409.60	3.640	23	850	74.34	5.150	4350	400	38	308.81		5.150	4350	400	58	471.35			
28	1.39	79.58	79°34'57"	5.191	5.110	4891	32	409.60	3.640	23	850	74.34	5.150	4350	400	38	308.81		5.150	4350	400	58	471.35			
29	1.35	77.25	77°15'1"	5.232	5.110	4932	32	412.84	3.640	23	850	74.34	5.250	4350	450	42	347.95		5.250	4350	450	56	463.93			
30	1.28	73.53	73°32'2"	5.472	5.260	5172	48	647.74	4.040	23	950	82.51	7.750	6350	700	76	929.44		7.750	6350	700	120	1,467.54			
					0.110	#DIV/0!									-25						-25					

dia.	linear wt.
10	0.617
12	0.888
16	1.578
20	2.466
25	3.853
29	5.185
32	6.313





C

Therefore:
c=t-150

BILL OF QUANTITIES PER TOWER

EXCAVATION	LEAN CONCRETE	REBAR	CONCRETE	BACKFILL
CU.M.	CU.M.	KGS	CU.M.	CU.M.
642.98	7.94	10,391.27	128.33	510.71
272.32	3.36	4,514.25	54.78	218.18
313.63	3.87	4,932.10	60.39	253.37
328.05	4.05	5,196.24	66.20	261.80
642.98	7.94	10,391.27	128.33	510.71
684.45	8.45	11,558.99	135.50	544.50
321.49	3.97	5,195.63	64.17	255.35
306.00	3.60	10,424.31	79.40	231.00
422.94	5.22	6,868.90	86.00	334.72
133.28	1.57	4,933.64	35.52	100.19
281.96	3.48	4,579.27	57.33	223.15
266.56	3.14	9,867.28	71.05	200.38
577.85	6.96	12,208.60	134.59	446.29
246.04	3.04	3,897.18	49.65	196.35
77.66	0.92	2,428.93	21.27	59.47
313.63	3.87	4,932.10	60.39	253.37
563.92	6.96	9,158.53	114.67	446.29
313.63	3.87	4,932.10	60.39	253.37
709.80	8.45	12,508.93	164.46	541.73
642.98	7.94	10,391.27	128.33	510.71
272.32	3.36	4,514.25	54.78	218.18
622.73	7.69	8,967.66	109.86	509.18
285.77	3.53	4,823.20	59.94	226.30
563.92	6.96	9,158.53	114.67	446.29
285.77	3.53	4,823.20	59.94	226.30
285.77	3.53	4,823.20	59.94	226.30
272.32	3.36	4,514.25	54.78	218.18
328.05	4.05	5,196.24	66.20	261.80
313.63	3.87	4,932.10	60.39	253.37
313.63	3.87	4,932.10	60.39	253.37
328.05	4.05	5,056.45	62.35	265.65
328.05	4.05	5,056.45	62.35	265.65
328.05	4.05	5,196.24	66.20	261.80
709.80	8.45	12,508.93	164.46	541.73
13,300.01 final	162.96 final	233,813.55 final	2,757.01 final	10,525.73 final